Three families from a $\mu = 19$ seed

Computed values, canonical corrections, and planar pairs

4 November 2026

Definition (coefficient). For $x < y$, put $\delta = \ell(y) - \ell(x)$. The ordinary KL coefficient is $\mu(x, y) = [q^{(\delta-1)/2}]P_{x,y}(q)$ when δ is odd, and zero otherwise. The computations below use the project's web/FKK convention: the readout is $[v^{-1}]$ of the corrected tensor coordinate, not the raw web coordinate. This v is the inverse of the positive-degree parameter.

Completed values. The three families share the same width-nine seed. The main family is also represented by a pair whose source has three six-boundary-point trees and additional tripods. Extensions A and B are different target extensions of a transported, tripod-source seed.

Width N	Main family	Extension A	Extension B	Status
9	19	19	19	Computed; seed agrees with W-graph
12	2446	598	598	Completed web-to-canonical reductions
15	0	0	0	Forced by parity
18	0	0	0	Forced by parity
21	unknown	unknown	unknown	No completed computation

The width-twelve main value was recovered for both the original-source and tripod-source representatives. This does not assert equality of their entire tensor coordinates or KL polynomials. The width-twelve values have not been independently checked by ordinary KL recursion.

Critical-coordinate audit at width twelve. All three rows use the tripod-source representative, with both tableaux promoted twice. Every nonzero critical correction was retained and audited; zero-contribution log entries were sampled, but not omitted from the computation. Each final offender queue was empty.

Family	Raw coefficient	Net correction	μ	Correction steps
Main	1 022 356	1 019 910	2446	8 245 220
A	86 689	86 091	598	1 747 417
B	87 205	86 607	598	1 872 327

These are elimination steps for the selected interval, not counts of nonzero terms in a full global basis expansion.

Guide to the planar pages. Pages 3–9 contain, in order: the original width-nine pair; its tripod-source representative; the original width-twelve pair; the exact rotated pair used overnight; the width-twelve tripod-source representative; Extension A; and Extension B. Each page includes both webs, both tableaux, permutation lengths, their difference, and the permitted critical degree. The drawing metadata and combinatorial graphs are saved beside the SVGs.

Growth status. The main values satisfy $2446 = 2^7 \cdot 19 + 14$ and $19/2^9 \approx 0.0371094$, $2446/2^{16} \approx 0.0373230$. Two nonzero values do not establish an asymptotic law. No unboundedness theorem is proved for these three candidates.

Definitions and the latest correction evidence

Definition (family words). Let $r \geq 0$ and $N = 9 + 3r$. A word records the row containing each entry of a standard tableau; A, B, C denote rows one, two, three. All words below have N occurrences of each letter and dominant prefixes. Use $a_r = (ABC)^N$ as source and one of the targets

$$\begin{aligned} g_r^{(0)} &= A^{5+3r}BA\,B^{5+3r}A^2CA\,C^{5+3r}B^2CBC^2, \\ g_r^{(A)} &= A^{5+3r}B^{1+3r}A\,B^5A^2C^{1+3r}A\,C^5B^2CBC^2, \\ g_r^{(B)} &= A^5BA\,B^5A^2CA^{1+3r}\,C^5B^{2+3r}C^{1+3r}BC^2. \end{aligned}$$

The original main pair, before transport to tripods, is

$$\Big(AAB\big((ABC)^{r+1}ABACBC\big)^2(ABC)^{r+1}BCC,\ A^NB^NC^N\Big).$$

The recorded simultaneous promotions and star operations preserve its coefficient and give the main tripod-source pair. The appended diagrams specify which representative is shown; representative changes can alter length differences.

Length check. Define $\pi(w)^{-1}$ by listing the positions of C, then B, then A, in increasing order within each group. For the unpromoted tripod-source pairs, the length differences are

$$\delta_j(r) = \frac{27r^2 + L_jr + 138}{2}, \qquad (L_0, L_A, L_B) = (117, 105, 69).$$

Indeed, $\ell(\pi(a_r)) = 3N(N+1)/2$ and $\ell(\pi(w)) = 3N^2 - \mathrm{inv}(w)$, where inversions use $A < B < C$. The target inversion counts are $39 + 18r$, $39 + 24r$, $39 + 42r$. Thus δ_j is even for $r \equiv 2, 3 \pmod 4$, proving the displayed zeros.

Finite verification (secondary cores). Among the patch-containing, noncanonical lower webs with nonzero critical contribution at width twelve, A gives 219 distinct cyclic cores and B gives 222. Their union has 435 cores; six are shared. Boundary order and the inherited cut are retained. All 435 global core reductions completed; all their lower terms passed the canonicality check, and only multipliers 1 and $d = v + v^{-1}$ occurred.

For a core c, take all induced hexagonal cycle red graphs and nice unions of elementary six-face rings. In a nice union no three selected faces meet at a vertex, and each selected face has internal dual degree two or three. Count each face set once; its level is the number of dual edges minus the number of selected faces. Let $R_j(c)$ sum the corresponding reduced webs at level j. Across these cores there are 640 level-zero and 55 level-one occurrences, with none at higher level. Exact comparison of all web coefficients verifies

$$\boxed{W_c - b_c = R_0(c) - (v + v^{-1})R_1(c)}.$$

The cycle sum alone works for 194 A cores and 192 B cores; the remaining 25 and 30 require the overlap subtraction. This statement concerns the listed secondary cores, not the whole target or omitted zero-critical terms.

Missing argument. Prove the overlap identity for open diagrams, control degrees after gluing, and prove closure under extension. A quadratic-exponential growth result also needs an increasing layer contribution or quadratically many controlled insertions. No such uniform identity or lower bound is established.

Records: `secondary-cores/audit.json`, `secondary-cores/linear-ring-summary.json`; definitions and run provenance: `../branch-experiments-2026-11-04.md`. Rebuild here with `pdflatex mu19-candidates-and-results.tex`.

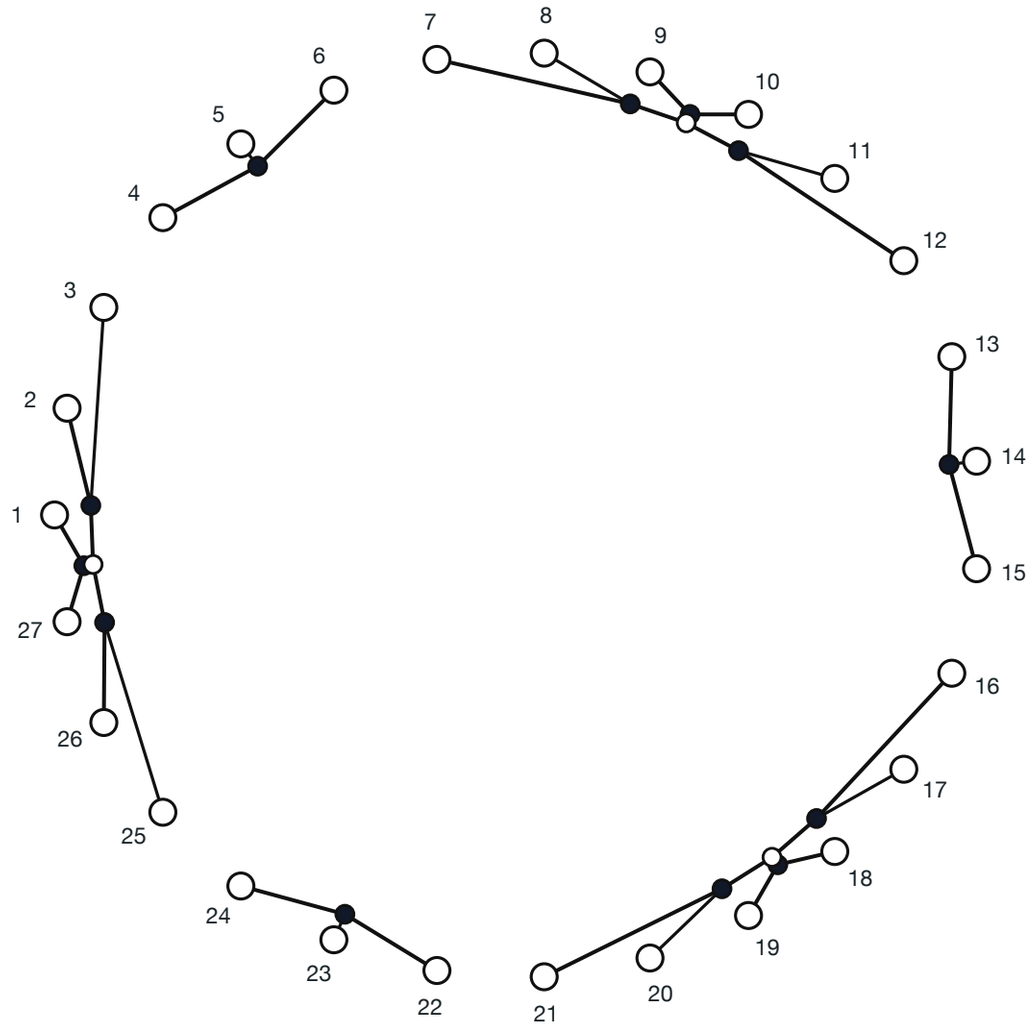
Original family: seed: (9,9,9) | μ = 19

Original source and target before promotion or star operations.

Original source: three six-point trees and three tripods.

Source planar web

Planar resolved web; fixed boundary labels

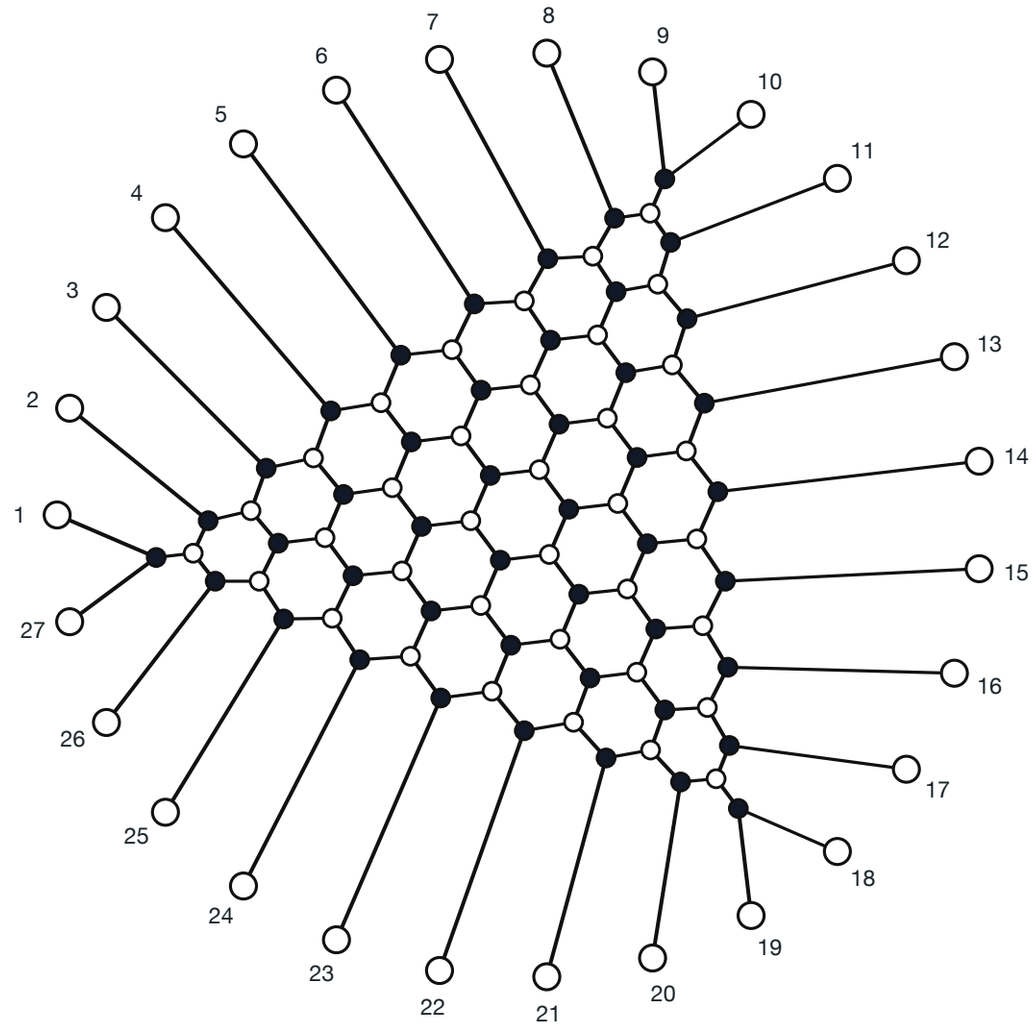


1	2	4	7	9	13	16	18	22
3	5	8	11	14	17	20	23	25
6	10	12	15	19	21	24	26	27

Permutation length: 168

Target planar web

Planar resolved web; fixed boundary labels



1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18
19	20	21	22	23	24	25	26	27

Permutation length: 243

C-B-A position convention. Length difference: 75. Permitted critical q-degree: 37.

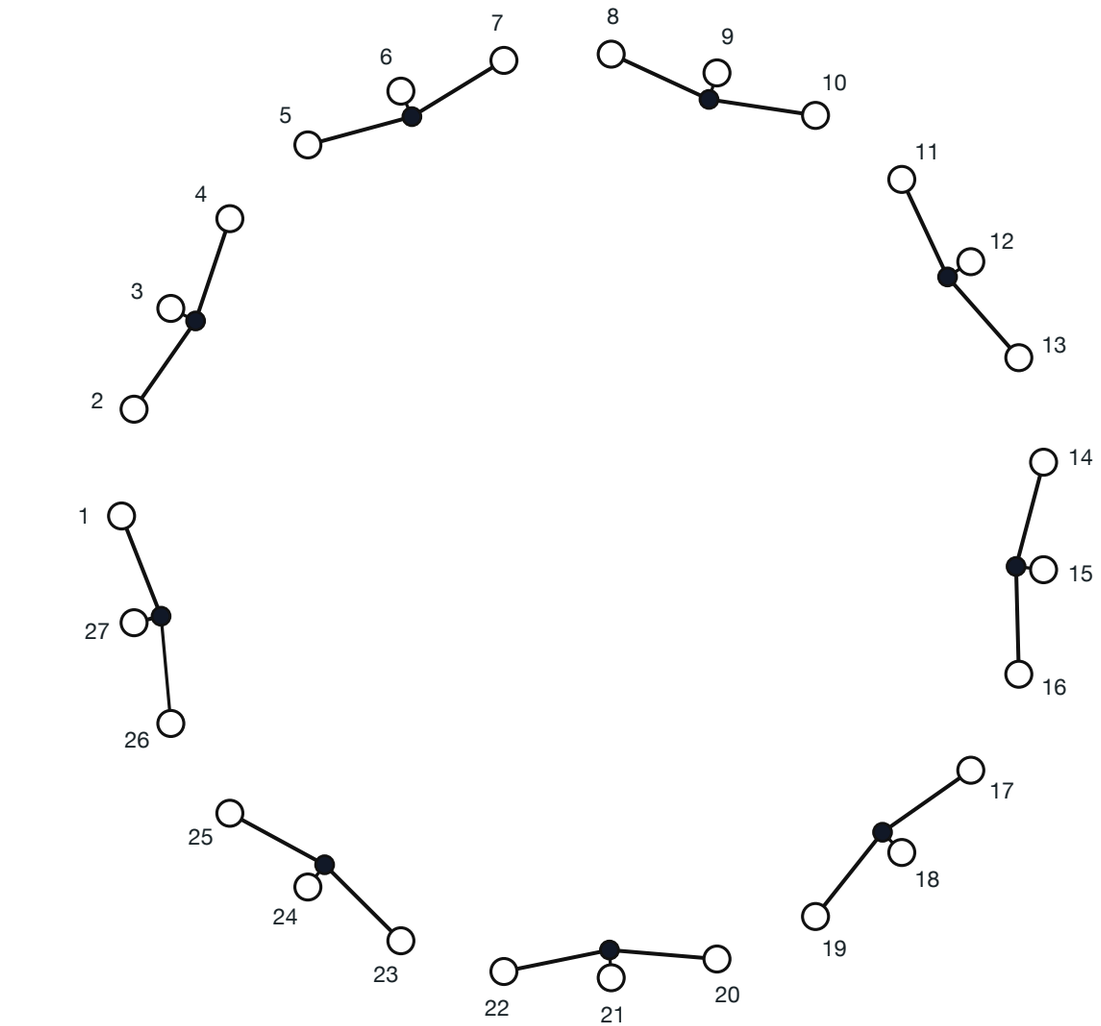
Tripod-source seed: (9,9,9) | μ = 19

Independent web reduction and catalog agree.

Two promotions of the tripod-source pair; the source is a union of tripods.

Source planar web

Planar resolved web; fixed boundary labels

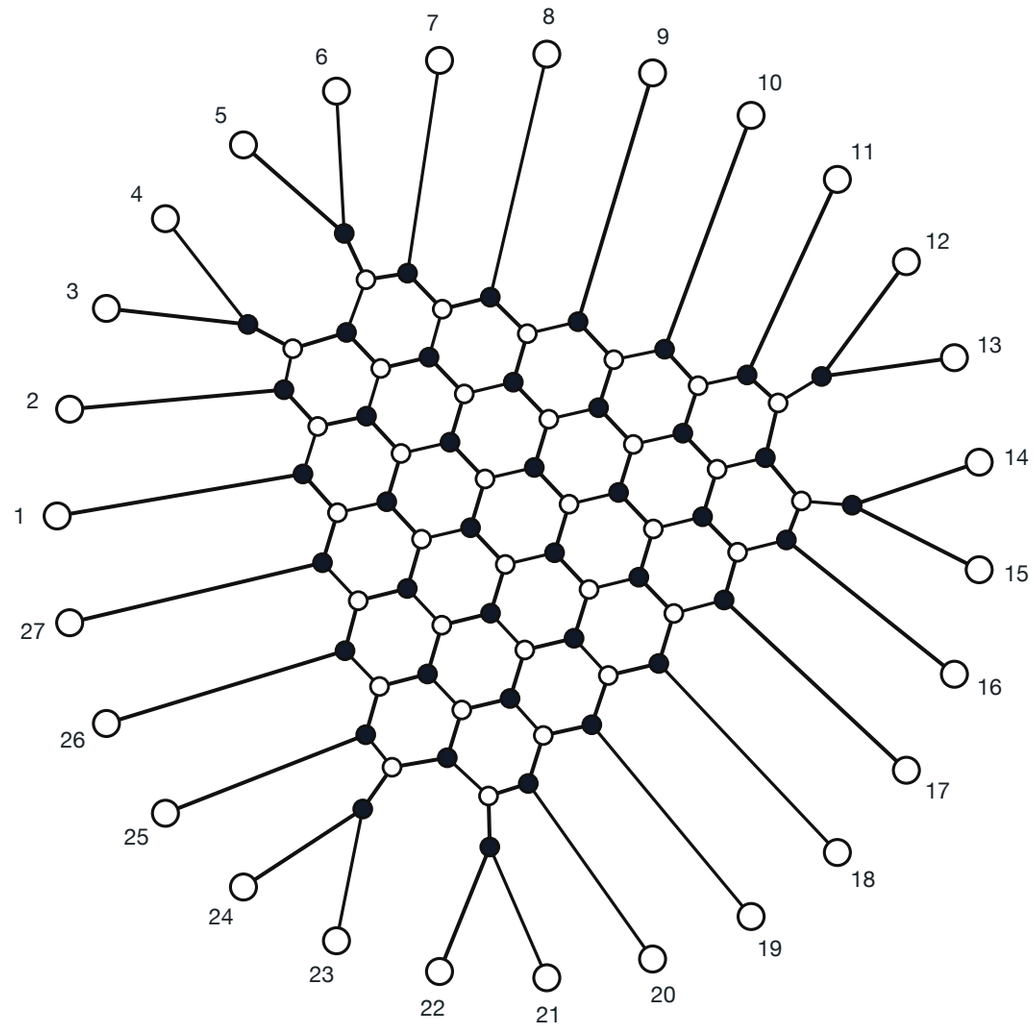


1	2	5	8	11	14	17	20	23
3	6	9	12	15	18	21	24	26
4	7	10	13	16	19	22	25	27

Permutation length: 151

Target planar web

Planar resolved web; fixed boundary labels



1	2	3	5	9	10	11	12	14
4	6	7	8	18	19	20	21	23
13	15	16	17	22	24	25	26	27

Permutation length: 200

C-B-A position convention. Length difference: 49. Permitted critical q-degree: 24.

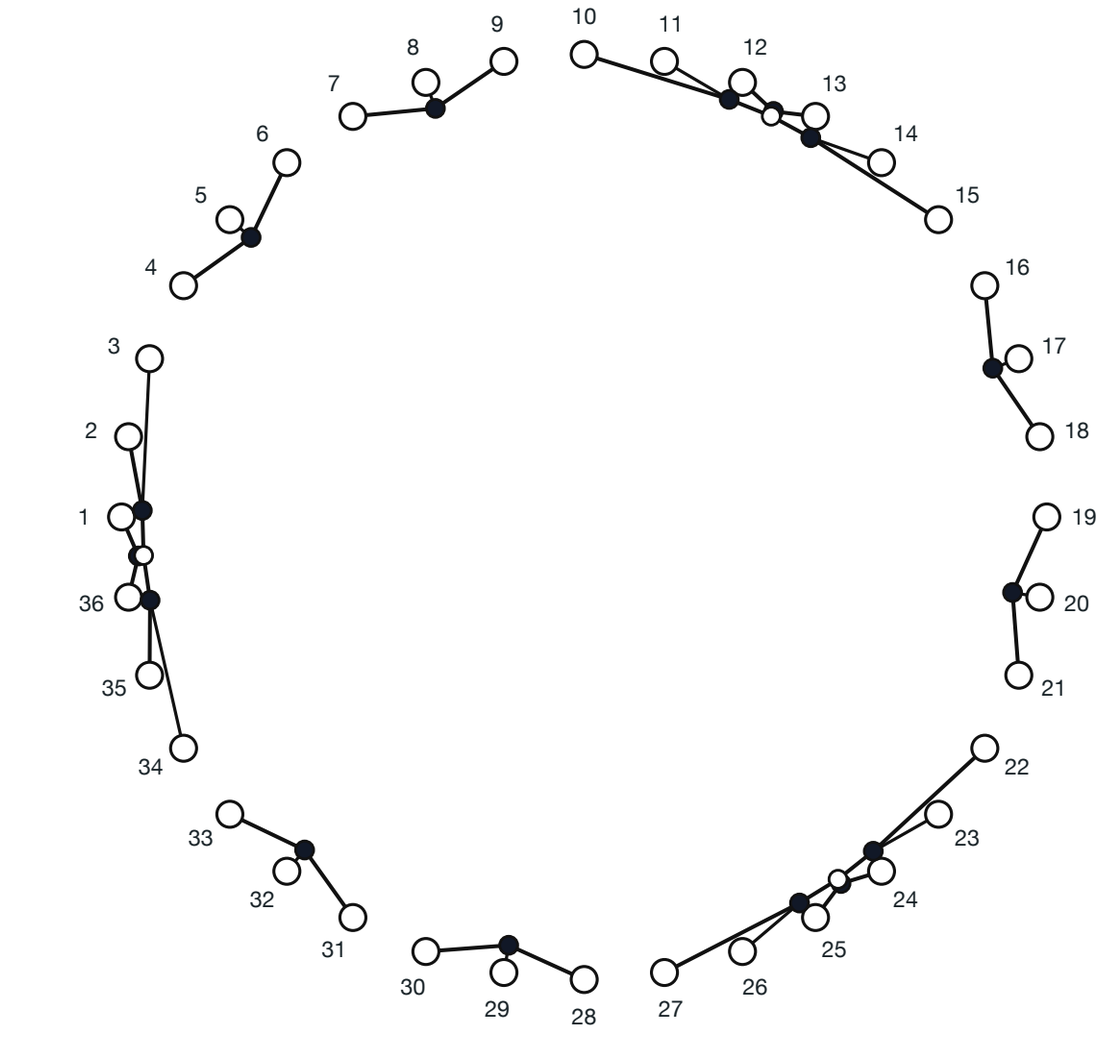
Original family: next member: (12,12,12) | μ = 2446

The completed overnight value belongs to this family; the run used the next page's rotated pair.

Original source: three six-point trees and six tripods. No star operations.

Source planar web

Planar resolved web; fixed boundary labels

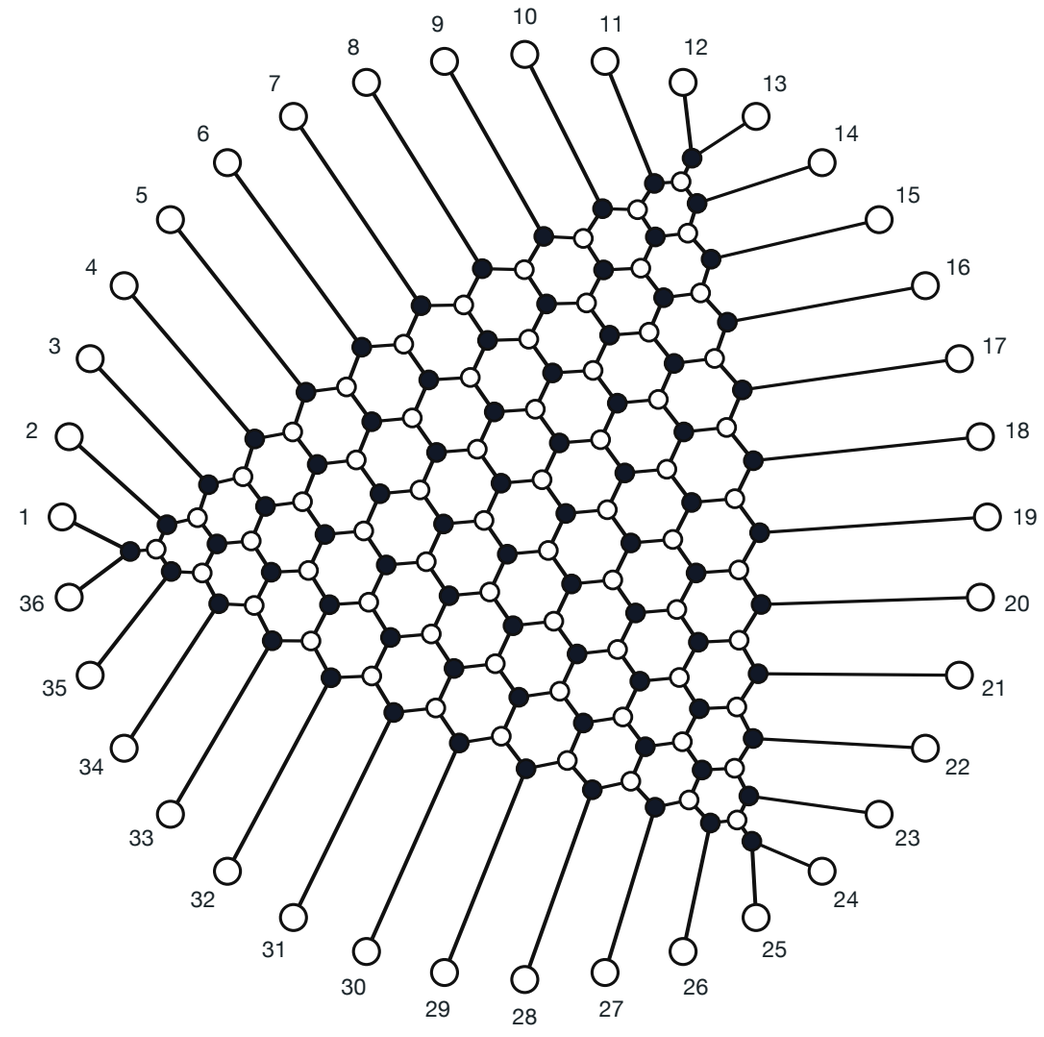


1	2	4	7	10	12	16	19	22	24	28	31
3	5	8	11	14	17	20	23	26	29	32	34
6	9	13	15	18	21	25	27	30	33	35	36

Permutation length: 279

Target planar web

Planar resolved web; fixed boundary labels



1	2	3	4	5	6	7	8	9	10	11	12
13	14	15	16	17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32	33	34	35	36

Permutation length: 432

C-B-A position convention. Length difference: 153. Permitted critical q-degree: 76.

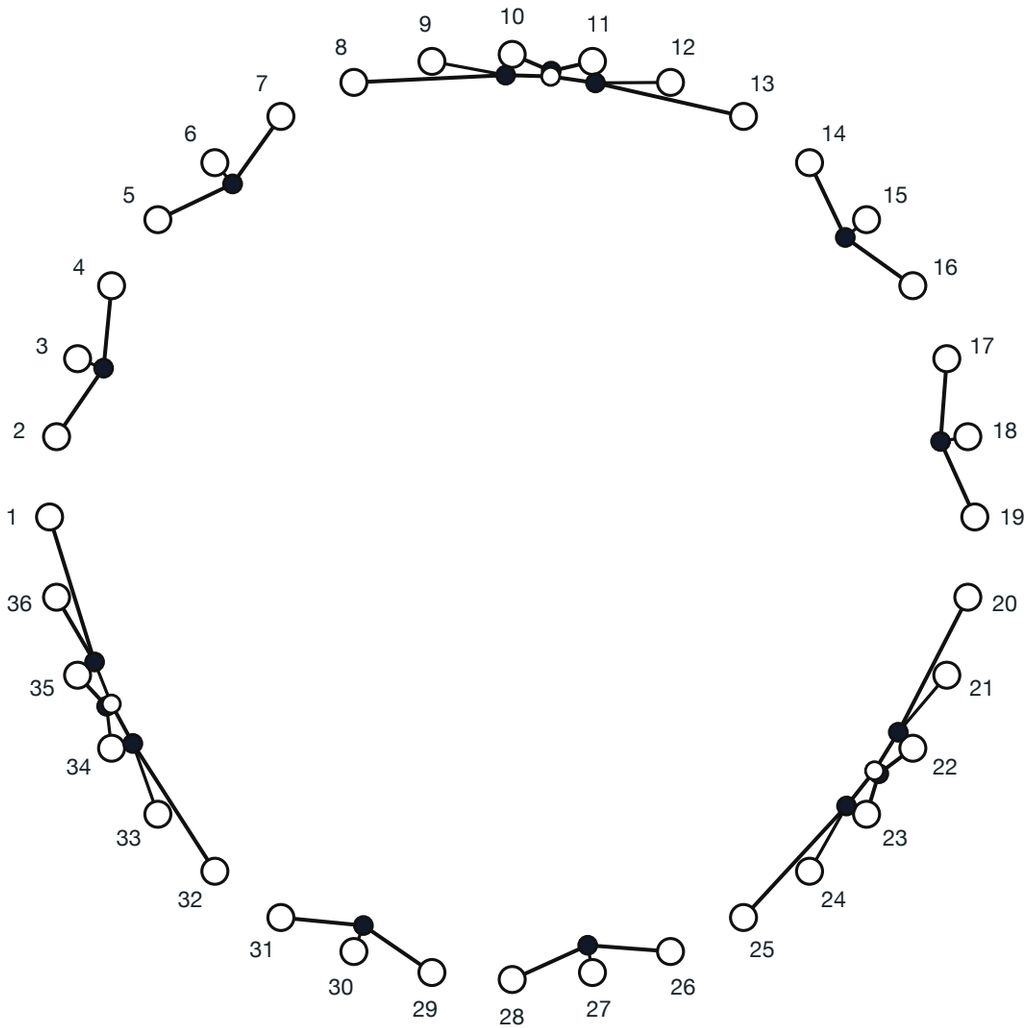
Exact overnight pair: (12,12,12) | μ = 2446

Copied from the completed balanced-k2 run record. This is NOT the tripod-source representative.

Two simultaneous promotions of the original pair; no star operations.

Source planar web

Planar resolved web; fixed boundary labels

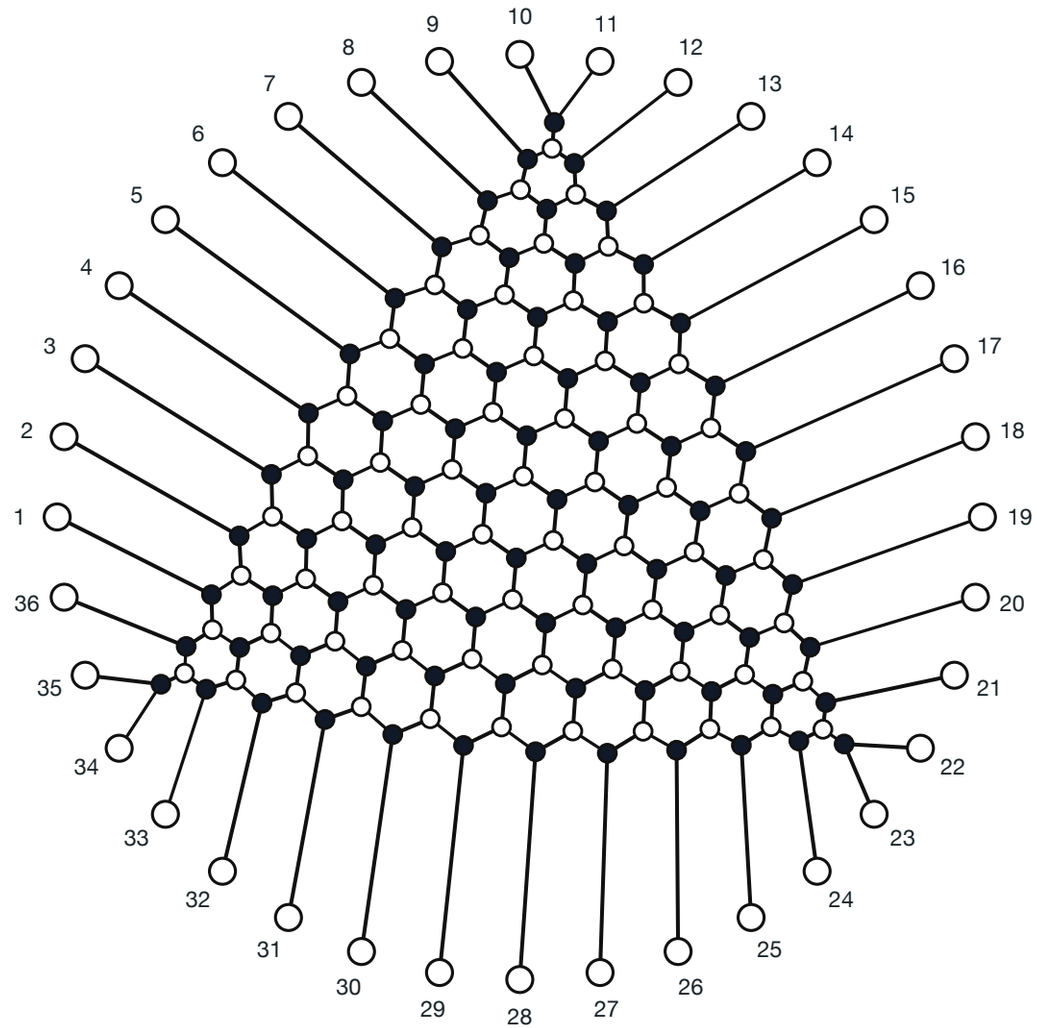


1	2	5	8	10	14	17	20	22	26	29	32
3	6	9	12	15	18	21	24	27	30	33	34
4	7	11	13	16	19	23	25	28	31	35	36

Permutation length: 259

Target planar web

Planar resolved web; fixed boundary labels



1	2	3	4	5	6	7	8	9	10	21	22
11	12	13	14	15	16	17	18	19	20	33	34
23	24	25	26	27	28	29	30	31	32	35	36

Permutation length: 392

C-B-A position convention. Length difference: 133. Permitted critical q-degree: 66.

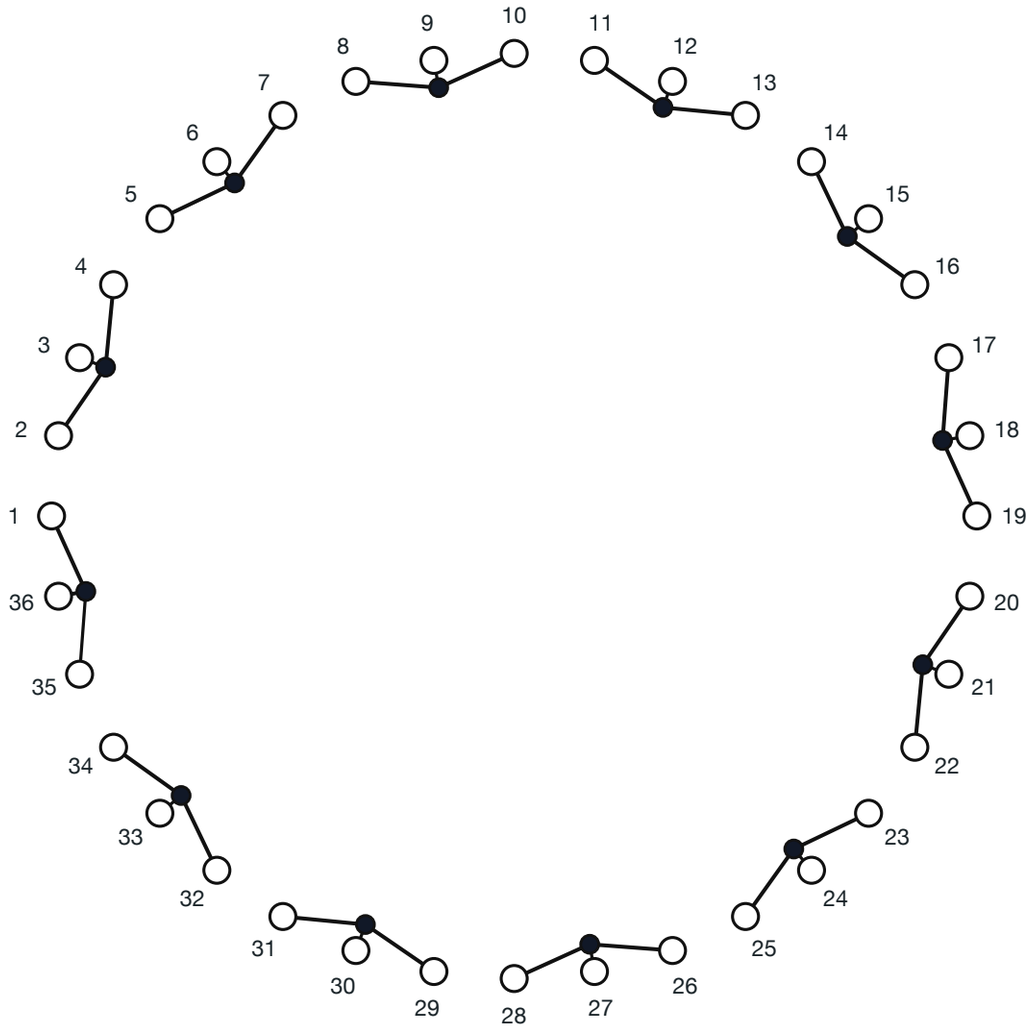
Original family: tripod representative: (12,12,12) | μ = 2446

Completed exact web-to-canonical reduction; final offender queue empty.

Two promotions of the tripod-source pair; the source is a union of tripods.

Source planar web

Planar resolved web; fixed boundary labels

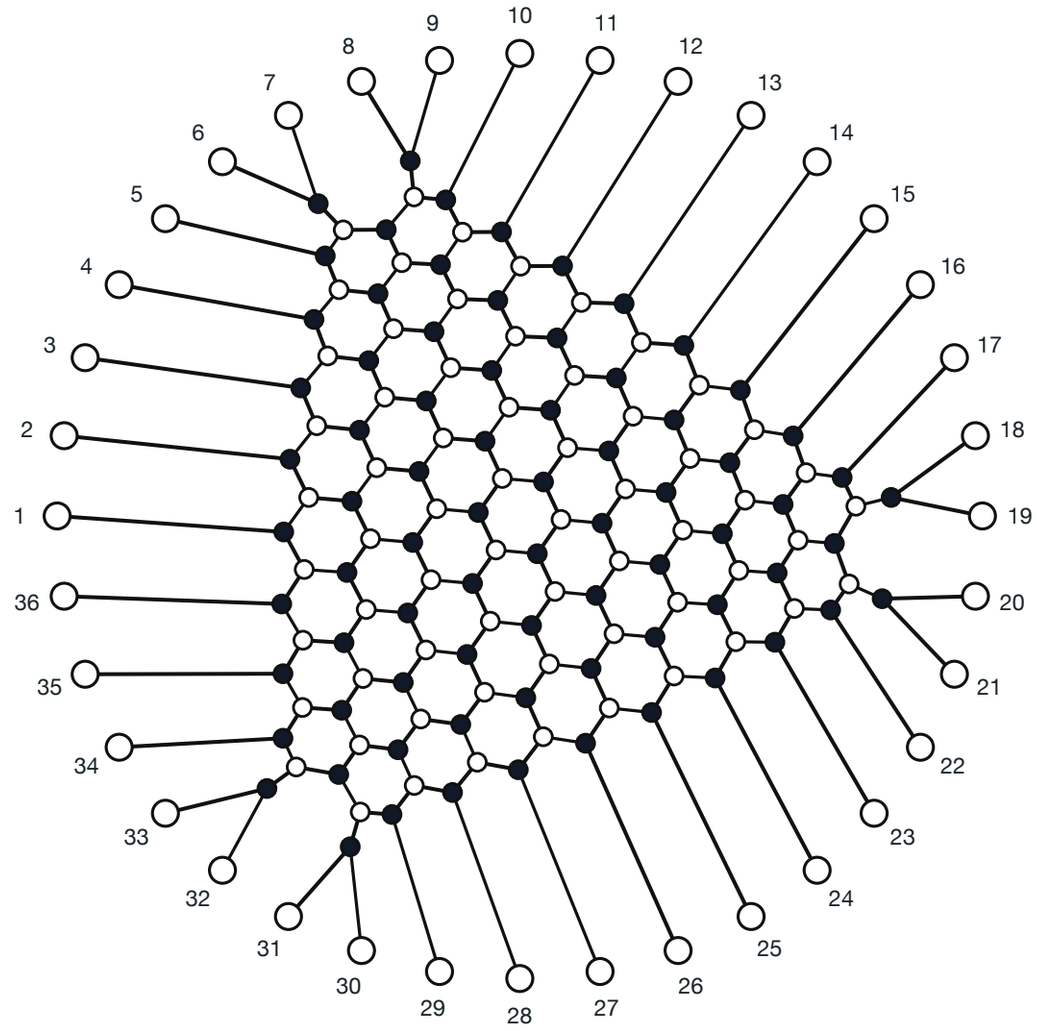


1	2	5	8	11	14	17	20	23	26	29	32
3	6	9	12	15	18	21	24	27	30	33	35
4	7	10	13	16	19	22	25	28	31	34	36

Permutation length: 256

Target planar web

Planar resolved web; fixed boundary labels



1	2	3	4	5	6	8	15	16	17	18	20
7	9	10	11	12	13	14	27	28	29	30	32
19	21	22	23	24	25	26	31	33	34	35	36

Permutation length: 359

C-B-A position convention. Length difference: 103. Permitted critical q-degree: 51.

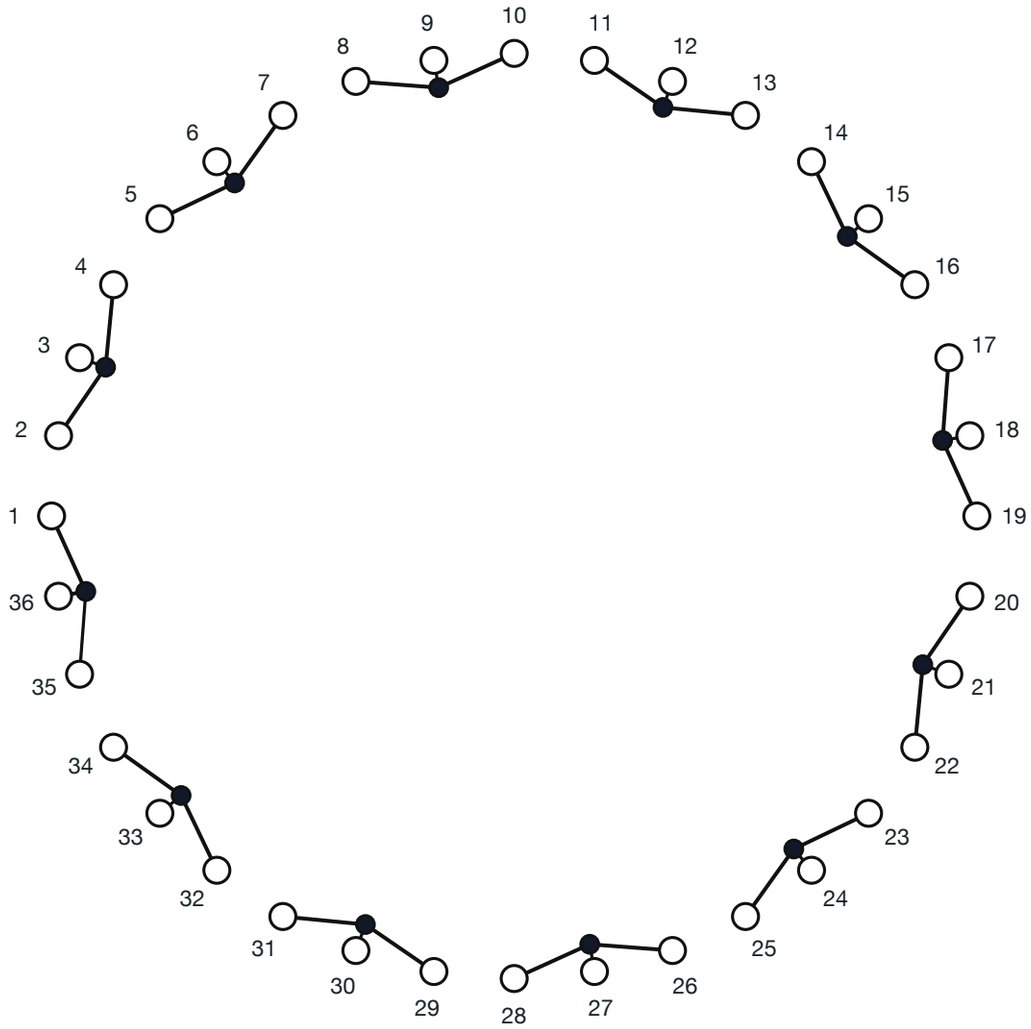
Extension A: (12,12,12) | μ = 598

Completed exact web-to-canonical reduction; final offender queue empty.

Two promotions of the tripod-source pair; the source is a union of tripods.

Source planar web

Planar resolved web; fixed boundary labels

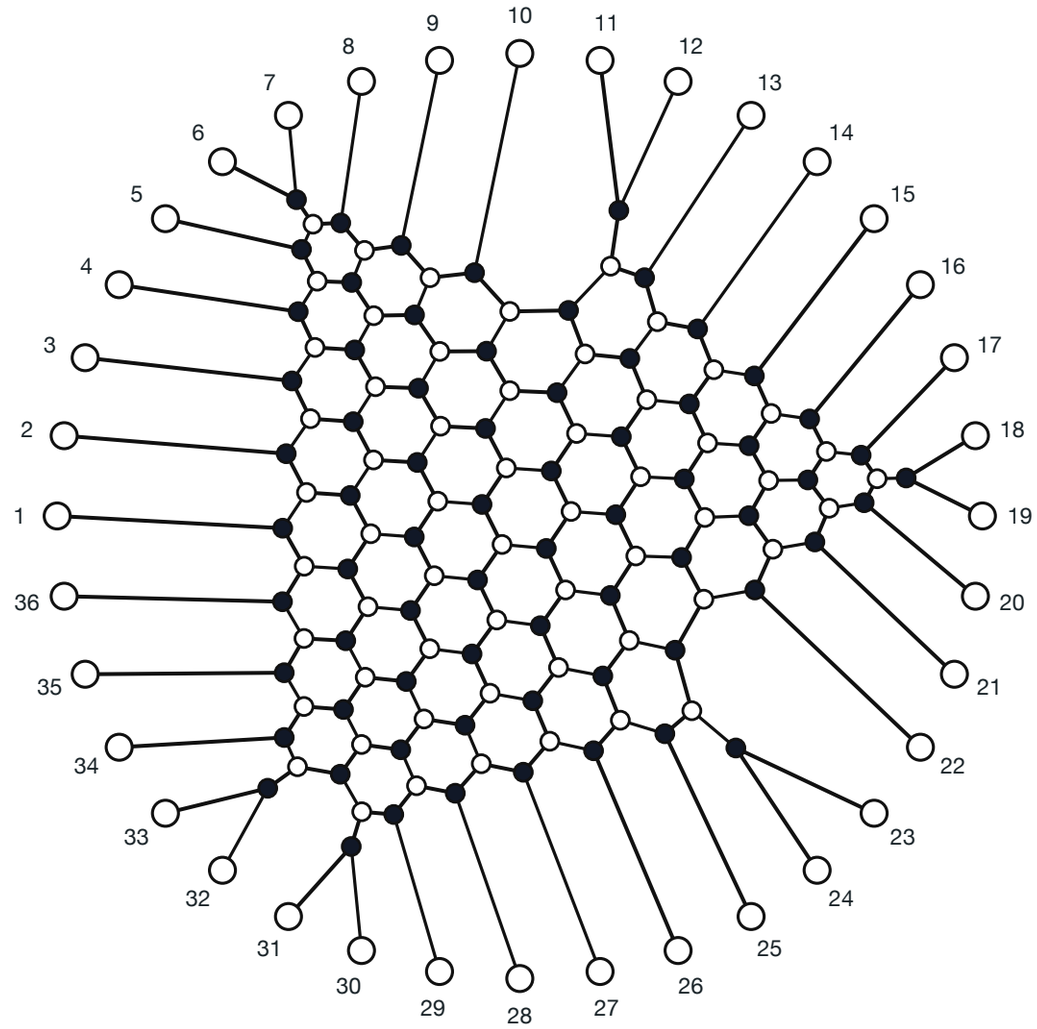


1	2	5	8	11	14	17	20	23	26	29	32
3	6	9	12	15	18	21	24	27	30	33	35
4	7	10	13	16	19	22	25	28	31	34	36

Permutation length: 256

Target planar web

Planar resolved web; fixed boundary labels



1	2	3	4	5	6	11	15	16	17	18	23
7	8	9	10	12	13	14	27	28	29	30	32
19	20	21	22	24	25	26	31	33	34	35	36

Permutation length: 353

C-B-A position convention. Length difference: 97. Permitted critical q-degree: 48.

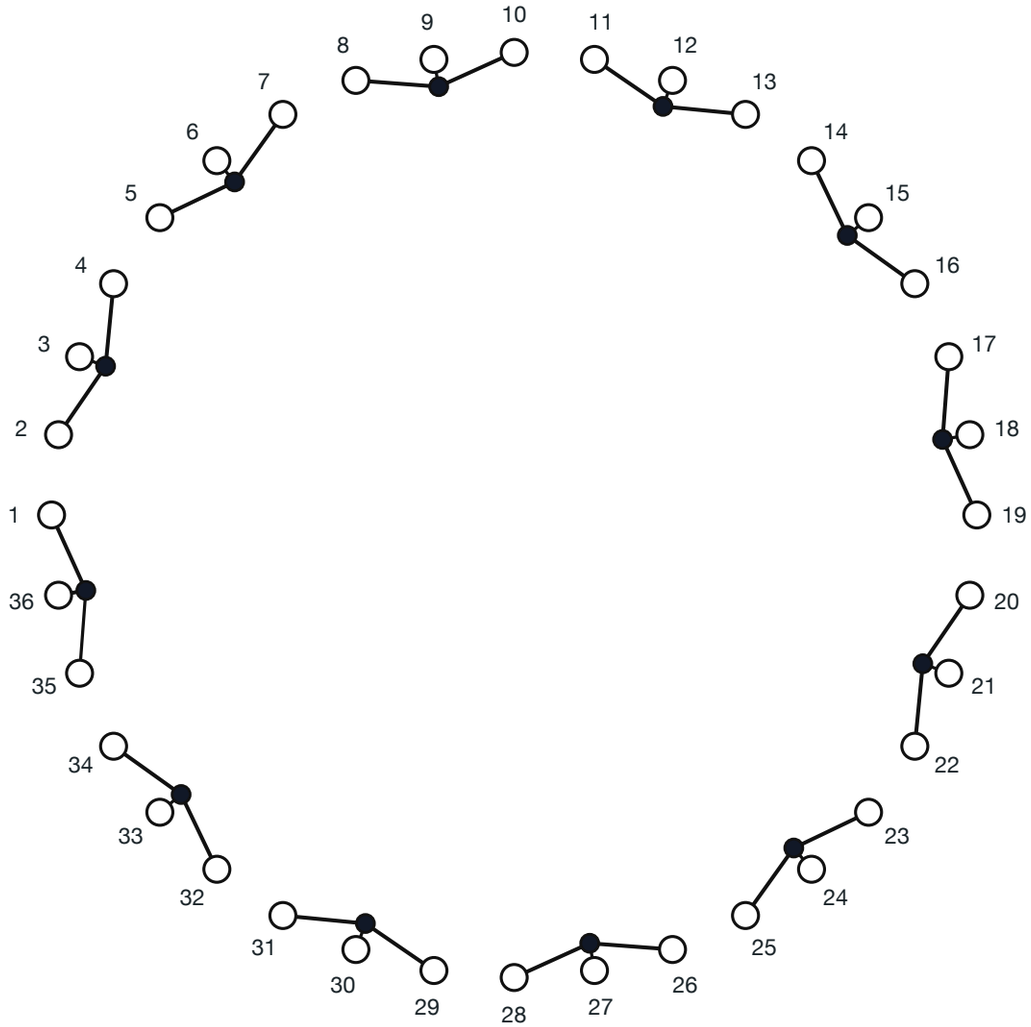
Extension B: (12,12,12) | μ = 598

Completed exact web-to-canonical reduction; final offender queue empty.

Two promotions of the tripod-source pair; the source is a union of tripods.

Source planar web

Planar resolved web; fixed boundary labels

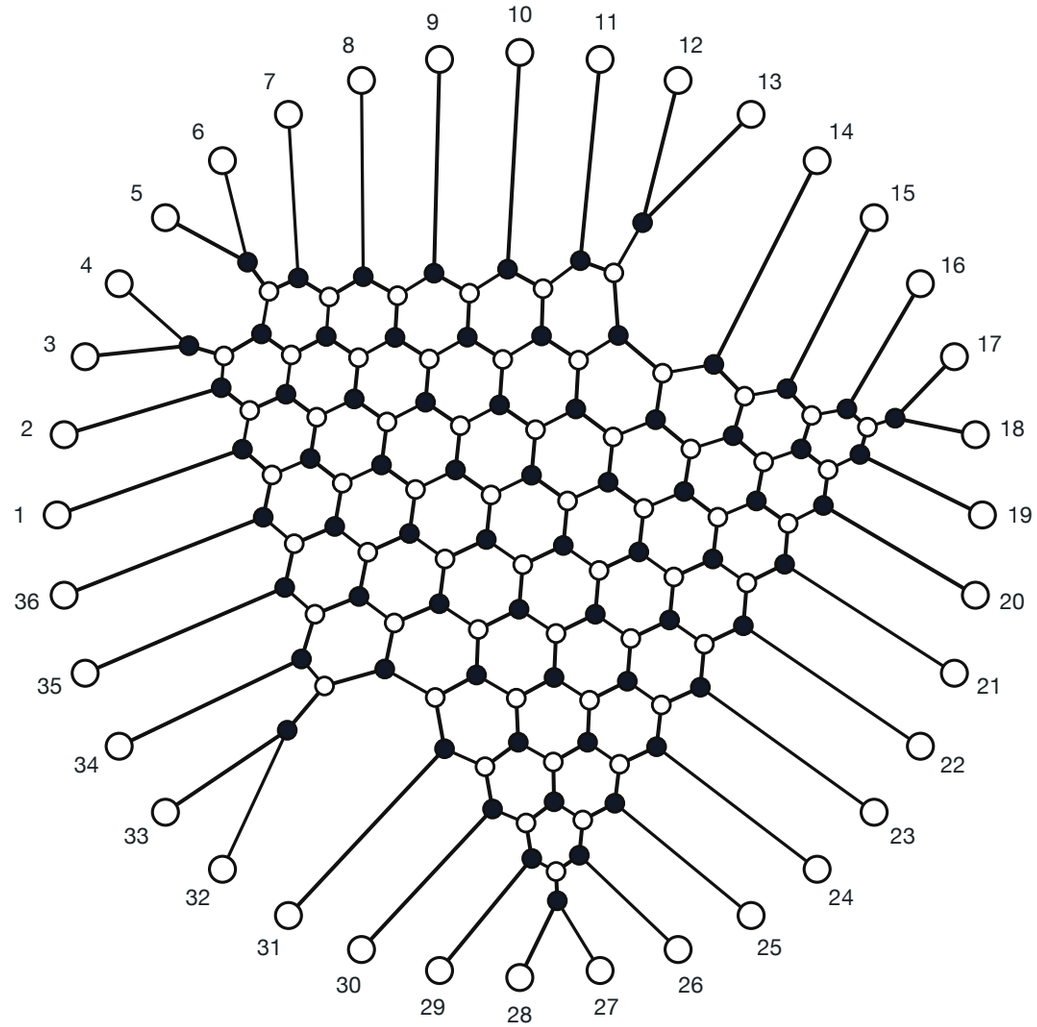


1	2	5	8	11	14	17	20	23	26	29	32
3	6	9	12	15	18	21	24	27	30	33	35
4	7	10	13	16	19	22	25	28	31	34	36

Permutation length: 256

Target planar web

Planar resolved web; fixed boundary labels



1	2	3	5	9	10	11	12	14	15	16	17
4	6	7	8	21	22	23	24	25	26	27	32
13	18	19	20	28	29	30	31	33	34	35	36

Permutation length: 359

C-B-A position convention. Length difference: 103. Permitted critical q-degree: 51.